Dress with gathered skirt · doll 45 cm (made to measure)

wide fit, Woven (0 % stretch) · original parametric draft. The pattern is yours to use.

> **Important:** With a new pattern, first sew a **test garment** from inexpensive fabric with similar stretch and check the fit.

Files

- PDF A4 tiles to glue (3 pages + overview): `doll_dress_45cm_a4_tiled.pdf`
- Preview A4 tiles: `doll_dress_45cm_a4_preview.png`
- SVG master 1:1: `doll_dress_45cm_master_1to1.svg`
- Cutting layout: `cutting_layout.png`
- Sewing instructions (PDF): `sewing_instructions.pdf`
- Sewing instructions (Markdown): `sewing_instructions.md`

Printing

- Always print at **100 % / "Actual size"** (not "Fit to page").
- Measure the **5 × 5 cm** control square before you cut.
- Plotter (610 mm roll): the page is 600 mm wide in portrait. In the print dialog choose a custom paper size with exactly this width and length, and turn rotation/scaling off.

Body measurements

Body measurement	Value	Source
Body height	45.0 cm	user
Chest	30.0 cm	user
Waist	28.0 cm	user
Hip	31.0 cm	user
Head	32.0 cm	user
Neck	15.1 cm	derived
Inseam	14.0 cm	user
Side length	22.0 cm	user
Arm length	12.0 cm	user
Back length	9.0 cm	derived
Shoulder width	3.2 cm	derived
Thigh	18.0 cm	derived
Ankle	11.4 cm	derived

Finished measurements (virtually sewn)

Finished measurement	Value
Chest width	36.9 cm
Hem width	60.9 cm
Dress length from neck point	20.7 cm
Sleeve length	4.2 cm
Sleeve opening	12.5 cm
Upper-arm width	12.5 cm
Neckline circumference	20.4 cm
Waist seam (bodice)	38.1 cm
Skirt top edge (ungathered)	60.9 cm
Skirt length	11.4 cm

Pattern pieces

Piece	Cutting	Fabric	Seam allowance
Front	cut 1 on fold	Main fabric	seam allowance 0.6 cm included; Neckline: 0.4 cm
Back	cut 2 (mirrored pair)	Main fabric	seam allowance 0.6 cm included; Back opening (turn under): 1.5 cm; Neckline: 0.4 cm
Sleeve	cut 2 (mirrored pair)	Main fabric	seam allowance 0.6 cm included; Sleeve hem: 1.0 cm
Neck binding (bias)	cut 1	Main fabric	seam allowance 0.4 cm included
Skirt front	cut 1 on fold	Main fabric	seam allowance 0.6 cm included; Hem: 1.0 cm
Skirt back	cut 2 (mirrored pair)	Main fabric	seam allowance 0.6 cm included; Hem: 1.0 cm; Back opening (turn under): 1.5 cm

Fabric needed

- **Main fabric:** about 0.25 m at 150 cm width (single layer, incl. 10 % for shrinkage; folded would need 0.35 m)
- Doll clothes: narrow seam allowances (6 mm). Sew with a short stitch (1.5–2 mm) and finish woven edges with pinking shears, a narrow zigzag or a little fray stop.
- Thin, soft fabrics work best (cotton poplin, batiste, old T-shirts); thick fabric makes the tiny seams bulky.
- The back opens completely: close it with narrow Velcro (1 cm) or 2–3 small snaps.

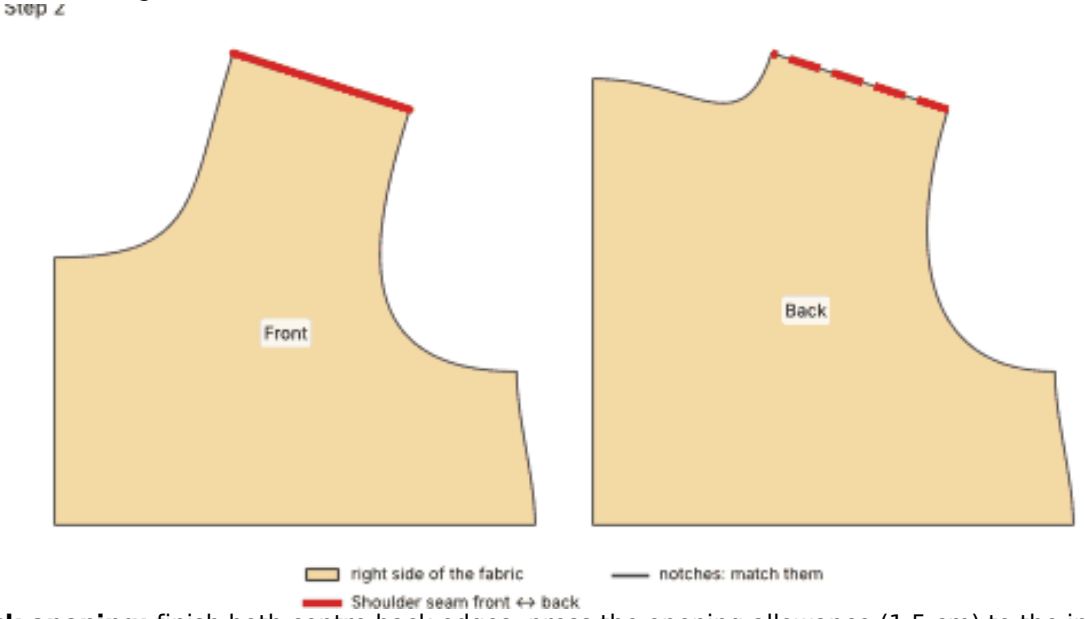
Cutting layout: see `cutting_layout.png` (grainline arrow parallel to the selvage, "flipped" = place the piece face down, "×2" = cut through both layers of the folded fabric, which gives a mirrored pair, "(unfolded)" = cut-on-fold piece shown opened out).

Cut on fold (Front, Skirt front): the paper piece is only one half. Its fold edge (dash-dot line, bracket arrows "PLACE ON FOLD") is the centre of the piece: it gets no seam allowance and is never cut. Where the cutting layout shows a piece "(unfolded)" on a single layer, fold the fabric along the grain just wide enough for the piece and place the fold edge on that fold, or trace the paper piece, flip it over along the fold edge and trace the second half.

Sewing instructions

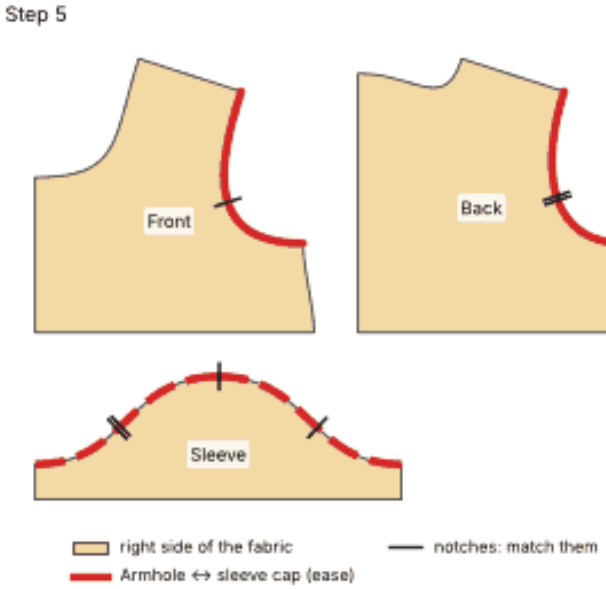
Seam allowance is included in the pattern (see table).

1. **Prepare:** pre-wash the fabric, cut out the pieces, snip the notches 3–4 mm deep.
2. **Shoulders:** place front and back right sides together and sew the shoulder seams with a straight stitch, then finish the edges.

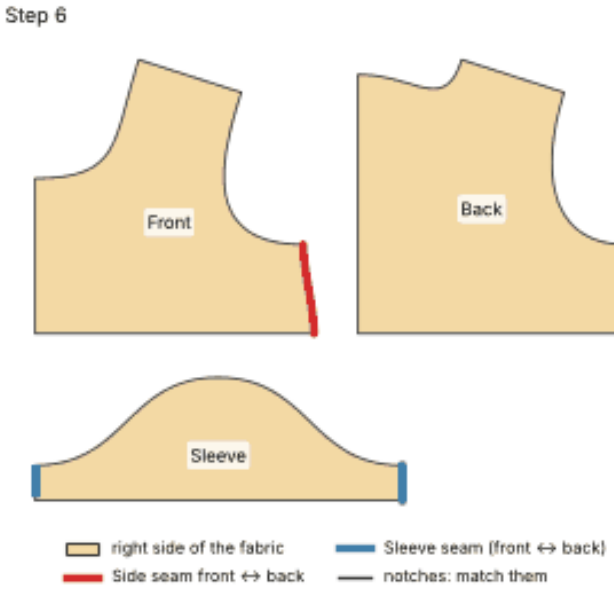


3. **Back opening:** finish both centre back edges, press the opening allowance (1.5 cm) to the inside and stitch it down. The two edges overlap at the centre back: sew a strip of narrow Velcro (or two or three snaps) onto the overlap, the soft side on the left back, the hook side on the right back. The garment opens completely, which makes dressing easy (dolls, small children).

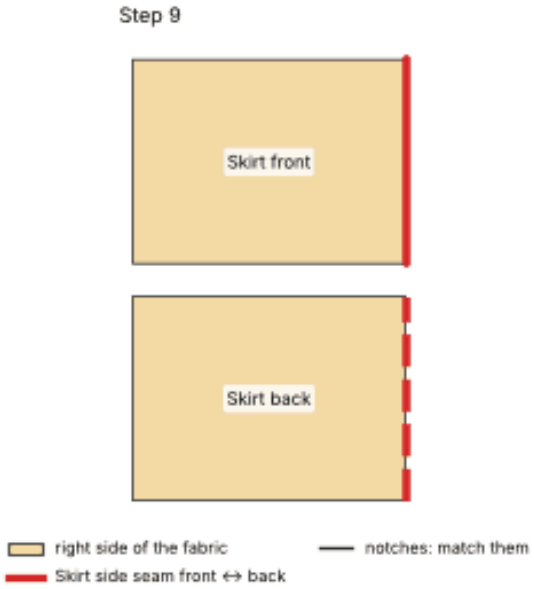
4. **Neck binding:** fold the bias strip lengthwise wrong sides together and press. Pin it right sides together to the neckline (raw edges together), letting each end stick out past the opening edge by the allowance, and sew. Fold the ends in, turn the binding to the inside so it covers the seam allowance, and stitch it down close to the fold.
5. **Set in the sleeves:** pin the centre of the sleeve cap (top notch) to the shoulder seam, match the front/back notches and sew the sleeve in flat (ease the cap in slightly if needed).



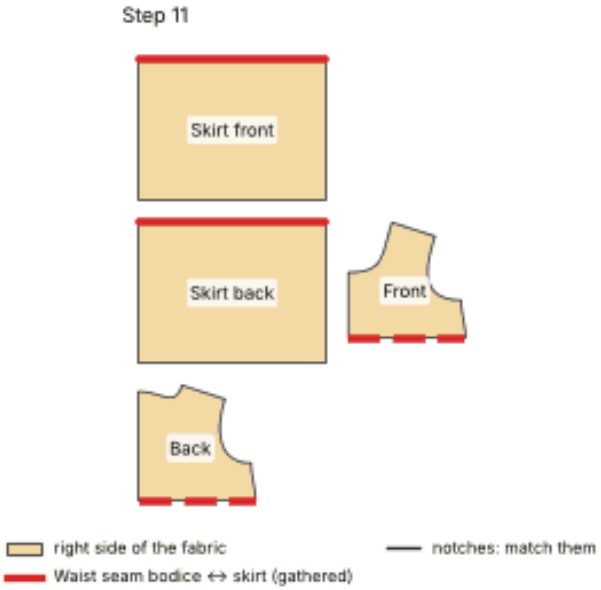
6. **Side and sleeve seams:** sew in one go from the hem over the underarm to the sleeve hem; the underarm seams meet exactly.



7. **Sleeve hem:** press the hem allowance under and sew with a straight stitch.
8. **Skirt centre back seam:** sew the skirt back pieces right sides together from the hem up to the notch; above the notch the seam stays open (it continues the back opening).
9. **Skirt side seams:** sew skirt front and back right sides together with a straight stitch, then finish the edges.



10. **Gather the skirt:** run two gathering threads along the top edge and pull them up to the bodice waist (gather ratio 1.6); spread the gathers evenly.
11. **Waist seam:** pin the skirt right sides together to the bodice (side seams meet, centre front/back meet) and sew with a straight stitch, then finish the edges.



12. **Hem:** press the hem allowance under twice and stitch.
13. Done! Press the seams.

Validation report

Sewability check

Area	Check	Target	Actual	Δ	Result	Note
Seam pairs	Side seam front ↔ back	3.1 cm	3.1 cm	+0.0 mm	✓ OK	
Seam pairs	Armhole ↔ sleeve cap (ease)	ease 3–8 mm	13.8 cm ↔ 14.3 cm	+5.4 mm	✓ OK	Woven: 3–8 mm
Seam pairs	Shoulder seam front ↔ back	3.7 cm	3.7 cm	+0.0 mm	✓ OK	
Seam pairs	Sleeve seam (front ↔ back)	1.2 cm	1.2 cm	+0.0 mm	✓ OK	
Seam pairs	Neck binding ↔ neckline	20.4 cm	20.4 cm	+0.0 mm	✓ OK	

Area	Check	Target	Actual	Δ	Result	Note
Seam pairs	Waist seam bodice ↔ skirt (gathered)	40–100 % of 60.9 cm	38.1 cm = 63 %	-228.4 mm	□ OK	gather ratio 1.6 (allowed 1–2.5)
Seam pairs	Skirt side seam front ↔ back	11.4 cm	11.4 cm	+0.0 mm	□ OK	
Seam pairs	Skirt centre back seam (left ↔ right)	7.4 cm	7.4 cm	+0.0 mm	□ OK	open for the top 4.0 cm (notch)
Angles/transitions	Underarm: side seam ↔ armhole front	90°	91.9°	+1.9°	□ OK	
Angles/transitions	Underarm: side seam ↔ armhole back	90°	92.2°	+2.2°	□ OK	
Angles/transitions	Shoulder ↔ armhole (front + back)	180°	180.6°	+0.6°	□ OK	
Angles/transitions	Sleeve cap top (smooth)	180°	178.2°	-1.8°	□ OK	
Angles/transitions	Sleeve underarm ↔ cap front	90°	90.9°	+0.9°	□ OK	
Angles/transitions	Shoulder ↔ neckline (front + back)	180°	180.9°	+0.9°	□ OK	
Angles/transitions	Neckline at centre front	90°	90.4°	+0.4°	□ OK	
Angles/transitions	Neckline at centre back	90°	89.0°	-1.0°	□ OK	
Angles/transitions	Hem at centre front	90°	90.0°	+0.0°	□ OK	
Angles/transitions	Hem ↔ side seam front	90°	88.8°	-1.2°	□ OK	
Angles/transitions	Skirt hem ↔ side seam	90°	90.0°	+0.0°	□ OK	
Angles/transitions	Skirt side seam ↔ waist	90°	90.0°	+0.0°	□ OK	
Angles/transitions	Skirt waist ↔ centre	90°	90.0°	+0.0°	□ OK	
Measurements	Chest circumference pattern = body + ease	36.6 cm	36.9 cm	+0.8 %	□ OK	body 30 cm + +22 % ease
Measurements	Head fits through the back opening	≥ 33.6 cm	38.1 cm	+45.0 mm	□ OK	neckline + both edges of the back opening ≥ head circumference + 5 %
Measurements	Dress goes on through the back opening	≥ 32.6 cm	46.0 cm	+134.9 mm	□ OK	waist seam + both edges of the skirt opening ≥ hip + 5 %
Measurements	Dress length plausible	14.9 cm – 33.5 cm	20.7 cm	+0.0 mm	□ OK	from the neck point
Geometry	Front: outline & seam allowance	closed, SA present	ok		□ OK	seam allowance 0.6 cm included; Neckline: 0.4 cm
Labels	Front: grainline & cutting	present	cut 1 on fold		□ OK	Main fabric
Geometry	Back: outline & seam allowance	closed, SA present	ok		□ OK	seam allowance 0.6 cm included; Back opening (turn under): 1.5 cm; Neckline: 0.4 cm
Labels	Back: grainline & cutting	present	cut 2 (mirrored pair)		□ OK	Main fabric

Area	Check	Target	Actual	Δ	Result	Note
Geometry	Sleeve: outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 0.6 cm included; Sleeve hem: 1.0 cm
Labels	Sleeve: grainline & cutting	present	cut 2 (mirrored pair)		✓ OK	Main fabric
Geometry	Neck binding (bias): outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 0.4 cm included
Labels	Neck binding (bias): grainline & cutting	present	cut 1		✓ OK	Main fabric
Geometry	Skirt front: outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 0.6 cm included; Hem: 1.0 cm
Labels	Skirt front: grainline & cutting	present	cut 1 on fold		✓ OK	Main fabric
Geometry	Skirt back: outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 0.6 cm included; Hem: 1.0 cm; Back opening (turn under): 1.5 cm
Labels	Skirt back: grainline & cutting	present	cut 2 (mirrored pair)		✓ OK	Main fabric
Geometry	Back: turned allowance lies flat	≤ 0.5 mm	0.0 mm		✓ OK	Back opening (turn under): corners mirrored at the fold line
Geometry	Sleeve: turned allowance lies flat	≤ 0.5 mm	0.0 mm		✓ OK	Sleeve hem: corners mirrored at the fold line
Geometry	Front: cut-on-fold edge	straight, no SA, 90° corners, on grain	ok, corners ≤ 0.4° off 90°		✓ OK	Centre (fold)
Geometry	Skirt front: cut-on-fold edge	straight, no SA, 90° corners, on grain	ok, corners ≤ 0.0° off 90°		✓ OK	Centre (fold)
Labels	Front: notches	on the cutting line	1 notch mark(s)		✓ OK	
Labels	Back: notches	on the cutting line	2 notch mark(s)		✓ OK	
Labels	Sleeve: notches	on the cutting line	4 notch mark(s)		✓ OK	1 at corners
Labels	Skirt back: notches	on the cutting line	2 notch mark(s)		✓ OK	

Result: 45 OK, 0 warning(s), 0 error(s): the pattern is sewable.

Construction values

Override any of these with the option `construction`, values in mm (e.g. `{"construction": {"chest_width": 366}}`).

Name	Value	Derived from	Note
`chest_width`	36.6 cm	chest × (1 + 22 % ease)	
`neck_width_half`	3.5 cm	neck / 5 + 1.5 cm	
`front_neck_depth`	4.0 cm	neck_width_half + 1.5 cm	
`back_neck_depth`	0.5 cm	fixed	
`shoulder_length`	3.7 cm	shoulder + allowance for fit 'wide'	
`shoulder_slope`	17	fixed	
`armhole_depth`	6.3 cm	18 % of chest_width + 2.5 cm + 3 %, limited to option armhole_depth_max	

Name	Value	Derived from	Note
body_length	9.3 cm	option body_length	

ting layout · Dress with gathered skirt doll 45 cm (made to meası

Main fabric: 150 cm wide, single layer
needed ≈ 0.25 m

